

Question number	Answer	Additional guidance	Mark
2(a)	A description that includes three of the following points: • fibrous (protein) (1) • composed of {three (polypeptide) chains in a helix / three-stranded helix / triple helix} (1) • held by hydrogen bonds between the chains (1) • credit details of the chains (1)	e.g. every third amino acid is glycine / repeating sequences of amino acids / high proportion of {glycine /	(3)

		proline / hydroxyproline}	
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Question number	Answer	Additional guidance	Mark
2(b)(i)	- tangent drawn correctly at 30 mins (1) - rate given (1)	ACCEPT straight line outside of curve at 30 mins ACCEPT in the range of 2 to 3.158 up to 3 dps Bald answer within the range = 2 marks Bald answer with too many dps = 1 mark	(2)

Question number	Answer	Additional guidance	Mark
2(b)(ii)	An explanation that includes three of the following points: <u>active site</u> (of protease) {binds to / fits} the {elastin / substrate} (1) - activation energy lowered (by the protease) (1) - breaking the <u>peptide</u> bonds (1) - by hydrolysis (1)	ACCEPT forms an enzyme-substrate complex DO NOT ACCEPT other named bonds dipeptide / polypeptide bonds	(3)